

Dr. Chilperic Armel Foko Kuate

Scientific Developer | Computational Tool Developer | Mathematical Modeller

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★ CORE CAPABILITIES

- Mechanistic & stochastic modelling
- Parameter estimation & optimisation
- Uncertainty & sensitivity analysis
- Scientific tools & reproducible workflows
- Computational tool development for science and education
- Cross-domain quantitative problem solving
- Technical communication, teaching & mentoring

</> TECHNICAL SKILLS

</> Languages

Python, Julia, R, MATLAB

⚙️ Optimisation

CasADi, IPOPT, CMA-ES, JuMP, Pyomo, MEALPY, pyMetaheuristic

📊 Scientific Computing

SciPy, NumPy, CVODE, SUNDIALS, lmfit

📈 Data & Visualisation

pandas, matplotlib, ggplot2, Seaborn, Plotly, Bokeh

🔍 Sensitivity Analysis

SALib — Sobol, Morris

🏠 Applied ML

regression, classification; scikit-learn

>_ Workflow

📄 Git, Jupyter, 🐧 Linux, HPC

🛠️ Tool Development

HTML, CSS, JavaScript, browser-based interfaces, interactive simulation tools

LANGUAGES

French Native

English Fluent

German B1+

Professional German in progress.

🏆 AWARDS & HONOURS

- ★ Bayer B4U Mentoring Programme, mentee (2026)
- ★ Marie Curie ITN Fellowship, PoLiMeR (2019–2023)
- ★ Post-AIMS Research Bursary (2017–2019)
- ★ Best Student, AIMS Ghana (2016)

👤 PROFESSIONAL SUMMARY

Scientific developer and computational tool developer with a research background in mathematical modelling, simulation, optimisation, and reproducible scientific computing across biological domains, from hepatic fatty-acid metabolism and immune-cell dynamics to plant photosynthetic adaptation across the C3 to C4 continuum. Skilled in parameter estimation, optimisation, uncertainty analysis, and scientific computing to translate complex biological questions into interpretable frameworks for hypothesis testing, experimental design, and model-guided decision-making. I build scientific web tools and simulation interfaces that make analysis workflows more usable for researchers, learners, and technical teams.

📁 PROFESSIONAL EXPERIENCE

Independent Research, Scientific Software, and Further Training Nov 2025 – Present *Düsseldorf, Germany*

- Advancing two manuscripts on mechanistic modeling of plant adaptation across the C3 to C4 continuum.
- Developing browser-based tools for scientific simulation and learning, including Foko Lab and Deutsch-WiPA 2026.
- Pursuing structured development in AI engineering and professional German to strengthen research software, technical implementation, and scientific work in German-speaking environments.

Postdoctoral Researcher Nov 2022 – Oct 2025 *CEPLAS / Institute of Computational Cell Biology, HHU Düsseldorf*

- Built a first-principles thermo-hydraulic-biochemical model of photosynthesis across the C3 to C4 continuum in *Flaveria*, integrating biochemical kinetics, heat transfer, hydraulics, and environmental forcing in one workflow.
- Combined Sobol sensitivity analysis, large simulation ensembles, and CMA-ES-based optimisation to reveal environment-specific adaptive trait combinations and prioritise candidate traits for crop engineering and model-guided experimental design.
- Used multiscale modeling to connect mesophyll and bundle-sheath physiology, transport constraints, and environmental response in plant adaptation.
- Presented this work at the CEPLAS Early Career Researcher Symposium, Düsseldorf, 2025.
- Supervised 2 MSc and 2 BSc projects in computational plant biology.

Marie Curie PhD Fellow Aug 2019 – Aug 2023 *Institute of Quantitative and Theoretical Biology, HHU Düsseldorf*

- Built mechanistic liver-metabolism models linking fatty-acid synthesis, oxidation, and degradation into interpretable dynamical systems for disease-relevant metabolic behavior.
- Established analytical conditions for bi-stability in fatty-acid metabolism and fitted semi-mechanistic models to time-course data, clarifying metabolic state switching and when reduced kinetics capture system behavior.
- Used parameter estimation, sensitivity analysis, and uncertainty quantification to strengthen robustness and interpretability in nonlinear biological models.
- Developed the enzyme-kinetic formalism for differentiable constraint-based metabolic models in a cross-institutional collaboration; published in *Metabolic Engineering* (2022).
- Compiled and formalised kinetic information for enzymes in animal fatty-acid synthesis; published as first author in *Bioscience Reports* (2023).

Research Fellow Nov 2017 – Mar 2019 *AIMS Cameroon & AIMS Rwanda*

- Developed stochastic and deterministic models of T-cell proliferation dynamics to extract interpretable parameters from CFSE-type experiments.
- Formulated a stochastic master-equation framework, derived analytical expressions for cell-count dynamics, and built a maximum-likelihood estimation workflow for model fitting.
- Worked across competing model classes, including ODE, delay, branching, and stochastic-process formulations, to assess biological interpretability and fitting behavior.

- ★ F.K.A. Allotey Meritorious Award (2016)
- ★ AIMS Ghana Hackathon Winner (2016)

🎓 EDUCATION

PhD, Quantitative & Theoretical Biology	
HHU Düsseldorf	2019–2023
Master 2 Mathematics & Applications	
Univ. Yaoundé I	2017–2019
MSc Mathematical Sciences	
AIMS Ghana	2015–2016
BSc Mathematics	
Univ. Douala	2009–2012

🔧 TEACHING & SUPERVISION

Supervised 2 MSc and 2 BSc projects in computational biology (HHU, 2019–2025).

Delivered 200+ teaching hours — mathematical modelling, ODEs, stochastic processes, Python (AIMS Rwanda, AIMS Cameroon, HHU).

💻 DEVELOPED COMPUTATIONAL TOOLS

Foko Lab

Browser-based environment for ODE, optimisation, steady-state, and stochastic modeling workflows.

🔗 [Open web tool](#)

Deutsch-WiPA 2026

Interactive B1 to B2 German learning tool with conjugation practice, feedback loops, and error tracking.

🔗 [Open web tool](#)

♥ INTERESTS

Agriculture and farming, CrossFit, cooking, music, and engagement with Survival International.

🎓 SELECTED TRAINING

🏆 Bayer B4U Mentoring Programme (Mentee)	Feb 2026
Inaugural edition in Germany; one-to-one pairing with Bayer scientists.	
</> AI Engineering Learning Track	2025–Present
Ongoing self-directed training in AI engineering, applied machine learning, and technical implementation workflows.	
🇩🇪 Professional German Development	Ongoing
Continued German progression beyond B1+ toward stronger professional fluency in academic and technical settings.	
🌱 PoLiMeR Marie Curie ITN Doctoral Training	2020–2022
Training across metabolism, regulation, enzyme catalysis, and scientific communication.	
💻 Advanced AI and Data Science Workshop	Dec 2021
HeiCAD, HHU Düsseldorf — neural networks, optimisation, and applied data science tools.	
🌱 Clinic on Meaningful Modelling of Epidemiological Data	Jul 2018
AIMS South Africa and SACEMA — epidemiological modeling and data-driven inference.	

💻 SELECTED COMPUTATIONAL TOOLS

- **Foko Lab**: browser-based modelling environment for ODE simulation, optimisation, steady-state analysis, stochastic models, and export workflows.
- **Deutsch-WiPA 2026**: B1–B2 professional German learning tool with vocabulary, grammar, conjugation practice, feedback loops, and progress-oriented training.
- **Research model repositories**: reproducible Python/R repositories for photosynthesis adaptation, fatty-acid metabolism, and T-cell proliferation models.

🎤 TALKS AND CONFERENCES

Talks

Oct 2025 CEPLAS Early Career Researcher Symposium, Düsseldorf, *talk*

Mar 2022 PoLiMeR Symposium, Düsseldorf, *talk & poster*

Jul 2019 Theoretical Biology Seminar, Univ. Rwanda

Aug 2018 Mathematics Against Malaria, Mit-tweida

Posters

Jul 2023 Economic Principles in Cell Biology Summer School

May 2022 Systems Biology of Mammalian Cells, Heidelberg

Jul 2020 International Systems Biology Group, Innsbruck

Jul 2018 MMED Epidemiological Modelling, Cape Town

📖 PUBLICATIONS

Published

- **Foko Kuate C.A.**, Ebenhöh O., Bakker B.M., Raguin A. *Kinetic Rate Laws of the Enzymes Involved in Animal Fatty Acid Synthesis. Bioscience Reports*, 2023.
- Besançon M., Dourado H., Kratochvíl M., **Foko Kuate C.A.**, Trefois C., Gu W., Ebenhöh O. *Interrogating the Effect of Enzyme Kinetics on Metabolism Using Differentiable Constraint-Based Models. Metabolic Engineering*, 2022.
- Dourado H., **Foko Kuate C.A.**, Liebermeister W., Lercher M.J. *Growth Mechanics and the Emergence of Metabolic Oscillations in Growing Cells. bioRxiv*, 2025.

+2 manuscripts in preparation on plant adaptation across the C3–C4 continuum.